Appendix

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Abstract

In this supplementary material, we begin by listing some lemmas required for our analysis, and then present a rigorous proof that Algorithm 1 converges to a KKT point.

Convergence of Algorithm 1

Lemma 1. *For the p-th power ($0 < p < 1$) of the Schatten p-norm, $\|\mathcal{A}\|_{\circledS p}^p$, its subgradient $\partial\|\mathcal{A}\|_{\circledS p}^p$ is bounded in the Frobenius norm. Specifically, there exists a constant $M > 0$ such that for any $G \in \partial\|\mathcal{A}\|_{\circledS p}^p$, we have $\|G\|_{\mathrm{F}} \leq M$.*

Proof Consider the t-SVD decomposition $\mathcal{A} = \mathcal{U} * \Sigma * \mathcal{V}^{\mathsf{T}}$, where Σ is the f-diagonal tensor of singular values $\sigma_i^{(j)} > 0$ for all i, j (since $\mathcal{A}$ has no zero singular values). For $0 < p < 1$, the subdifferential of $\|\mathcal{A}\|_{\circledS p}^p$ at a point with positive singular values consists of a single element:

$$\partial\|\mathcal{A}\|_{\circledS p}^p = \left\{\mathcal{U} * \mathcal{D} * \mathcal{V}^{\mathsf{T}}\right\}$$

where $\mathcal{D}$ is an f-diagonal tensor with diagonal elements:

$$\mathcal{D}_{ii}^{(j)} = p(\sigma_i^{(j)})^{p-1}$$

The Frobenius norm of this subgradient is bounded as follows:

$$\begin{aligned}
\|G\|_{\mathrm{F}} &= \|\mathcal{U} * \mathcal{D} * \mathcal{V}^{\mathsf{T}}\|_{\mathrm{F}} \\
&= \left(\frac{1}{n_3}\sum_{j=1}^{n_3} \|\bar{U}^{(j)}\bar{D}^{(j)}\bar{V}^{(j)\mathsf{T}}\|_{\mathrm{F}}^2\right)^{1/2} \\
&= \left(\frac{1}{n_3}\sum_{j=1}^{n_3} \|\bar{D}^{(j)}\|_{\mathrm{F}}^2\right)^{1/2} \qquad \text{(by unitary invariance)} \\
&= \left(\frac{1}{n_3}\sum_{j=1}^{n_3}\sum_{i=1}^{\min(n_1,n_2)} p^2(\sigma_i^{(j)})^{2(p-1)}\right)^{1/2}
\end{aligned}$$

Since $\mathcal{A}$ is fixed with no zero singular values, there exists $\sigma_{\min} > 0$ such that:

$$\sigma_i^{(j)} \geq \sigma_{\min} > 0 \quad \forall i, j$$

Because $0 < p < 1$, we have $p - 1 < 0$, so:

$$(\sigma_i^{(j)})^{2(p-1)} \leq (\sigma_{\min})^{2(p-1)}$$

Thus the norm is bounded by:

$$\begin{aligned}
\|G\|_{\mathrm{F}} &\leq \left(\frac{1}{n_3}\sum_{j=1}^{n_3}\sum_{i=1}^{\min(n_1,n_2)} p^2(\sigma_{\min})^{2(p-1)}\right)^{1/2} \\
&= p(\sigma_{\min})^{p-1}\sqrt{\frac{1}{n_3}\sum_{j=1}^{n_3}\min(n_1,n_2)} \\
&= p(\sigma_{\min})^{p-1}\sqrt{\min(n_1,n_2)}
\end{aligned}$$

Since $\sigma_{\min} > 0$ is fixed and depends only on $\mathcal{A}$, and p is fixed with $0 < p < 1$, this expression is bounded. Therefore, there exists $M > 0$ such that $\|G\|_{\mathrm{F}} \leq M$.

Lemma 2. *Let $\mathcal{A}$ be a third-order tensor with finite Frobenius norm, i.e., $\|\mathcal{A}\|_{\mathrm{F}} \leq C$ for some constant $C > 0$. Then for $k \in \{1, 2, 3\}$, both the k-mode gradient operator $\nabla_k(\cdot)$ and its adjoint $\nabla_k^{\mathsf{T}}(\cdot)$ are bounded linear operators satisfying*

$$\|\nabla_k(\mathcal{A})\|_{\mathrm{F}} \leq C \quad \textit{and} \quad \|\nabla_k^{\mathsf{T}}(\mathcal{A})\|_{\mathrm{F}} \leq C.$$

Proof Let $\mathcal{A} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ satisfy $\|\mathcal{A}\|_{\mathrm{F}} \leq C$ for some constant $C > 0$. For any $k \in \{1, 2, 3\}$, the k-mode gradient operator $\nabla_k(\cdot)$ is defined as:

$$\nabla_k(\mathcal{A}) = \mathcal{G}_k = \mathcal{A} \times_k \mathbf{D}_{n_k},$$

where $\mathbf{D}_{n_k}$ is the circulant matrix generated by $\mathbf{d} = [-1, 1, 0, \ldots, 0]^{\mathsf{T}} \in \mathbb{R}^{n_k}$. By Definition 2 (mode-$k$ product), the k-mode unfolding gives:

$$(\mathcal{G}_k)_{(k)} = \mathbf{D}_{n_k}\mathcal{A}_{(k)}.$$

The Frobenius norm satisfies:

$$\|\nabla_k(\mathcal{A})\|_{\mathrm{F}} = \|(\mathcal{G}_k)_{(k)}\|_{\mathrm{F}} = \|\mathbf{D}_{n_k}\mathcal{A}_{(k)}\|_{\mathrm{F}}.$$

Using the submultiplicative property of matrix norms:

$$\|\mathbf{D}_{n_k}\mathcal{A}_{(k)}\|_{\mathrm{F}} \leq \|\mathbf{D}_{n_k}\|_2 \cdot \|\mathcal{A}_{(k)}\|_{\mathrm{F}},$$

where $\|\cdot\|_2$ is the spectral norm. Since $\|\mathcal{A}_{(k)}\|_{\mathrm{F}} = \|\mathcal{A}\|_{\mathrm{F}} \leq C$, it suffices to bound $\|\mathbf{D}_{n_k}\|_2$.

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The circulant matrix $\mathbf{D}_{n_k}$ has eigenvalues: $\lambda_j = -1 + \exp\left(-\frac{2\pi ij}{n_k}\right) = -1 + \cos\theta_j - i\sin\theta_j$, $\theta_j = \frac{2\pi j}{n_k}$, $j = 0, \ldots, n_k - 1$. The magnitude is: $|\lambda_j| = \sqrt{(-1+\cos\theta_j)^2 + (-\sin\theta_j)^2} = \sqrt{2 - 2\cos\theta_j} = 2\left|\sin\left(\frac{\pi j}{n_k}\right)\right|$. Since $|\sin x| \leq 1$ for all real x, we have $\max_j |\lambda_j| \leq 2$. Thus:

$$\|\mathbf{D}_{n_k}\|_2 = \max_j |\lambda_j| \leq 2.$$

This implies:

$$\|\nabla_k(\mathcal{A})\|_{\mathrm{F}} \leq 2\|\mathcal{A}\|_{\mathrm{F}} \leq 2C.$$

For the adjoint operator $\nabla_k^{\mathsf{T}}(\mathcal{A})$, Definition 2 gives:

$$\nabla_k^{\mathsf{T}}(\mathcal{A}) = \mathcal{A} \times_k \mathbf{D}_{n_k}^{\mathsf{T}},$$

with k-mode unfolding $\mathbf{D}_{n_k}^{\mathsf{T}}\mathcal{A}_{(k)}$. Since $\|\mathbf{D}_{n_k}^{\mathsf{T}}\|_2 = \|\mathbf{D}_{n_k}\|_2 \leq 2$:

$$\|\nabla_k^{\mathsf{T}}(\mathcal{A})\|_{\mathrm{F}} \leq 2\|\mathcal{A}\|_{\mathrm{F}} \leq 2C.$$

The lemma bounds hold with constant $2C$ instead of C, but since $C > 0$ is arbitrary, the operators remain bounded. To match the lemma statement exactly, redefine $\widetilde{C} = 2C$. Then $\|\mathcal{A}\|_{\mathrm{F}} \leq \widetilde{C}/2$ implies both gradient norms are $\leq \widetilde{C}$.

Theorem 1 (Convergence Analysis of Algorithm 1)**.** *Let $P_t = \{\mathcal{Z}_t, \mathbf{E}_{1,t}^v, \mathbf{E}_{2,t}^v, \mathbf{J}_t^v, \mathbf{O}_t^v, \mathcal{N}_t^k, \mathcal{G}_{k,t}, \mathcal{C}_t, \mathcal{A}_t\}$ denote the iteration sequence generated by Algorithm 1. Then:*

1. *$\{P_t\}$ is bounded;*
2. *Any accumulation point of Algorithm 1 is a KKT point.*

Proof of (1). ① For the $\mathbf{E}_{1,t+1}^v$ subproblem, the first-order optimality condition yields

$$0 \in \partial\lambda_2\|\mathbf{E}_{1,t+1}^v\|_{2,1} - \mathbf{J}_t^v - \mu_t(\mathbf{X}^v - \mathbf{X}^v\mathbf{Z}_{t+1}^v - \mathbf{E}_{1,t+1}^v). \tag{1}$$

Since $\mathbf{J}_{t+1}^v = \mathbf{J}_t^v + \mu_t(\mathbf{X}^v - \mathbf{X}^v\mathbf{Z}_{t+1}^v - \mathbf{E}_{1,t+1}^v)$, it follows that

$$0 \in \partial\lambda_2\|\mathbf{E}_{1,t+1}^v\|_{2,1} - \mathbf{J}_{t+1}^v. \tag{2}$$

This implies

$$\mathbf{J}_{t+1}^v(:,i) \in \partial\lambda_2\|\mathbf{E}_{1,t+1}^v(:,i)\|_2, \tag{3}$$

and by the subgradient definition

$$\partial\|\mathbf{E}_{1,t+1}^v(:,i)\|_2 = \begin{cases} \dfrac{\mathbf{E}_{1,t+1}^v(:,i)}{\left\|\mathbf{E}_{1,t+1}^v(:,i)\right\|_2}, & \mathbf{E}_{1,t+1}^v(:,i) \neq \mathbf{0} \\ \{g \mid \|g\|_2 \leq 1\}, & \mathbf{E}_{1,t+1}^v(:,i) = \mathbf{0} \end{cases}, \tag{4}$$

we have $\|\mathbf{J}_{t+1}^v(:,i)\|_2 \leq \lambda_2$. Thus $\{\mathbf{J}_t^v\}$ is bounded. Similarly, for the $\mathbf{E}_{2,t+1}^v$ subproblem, using

$$\mathbf{O}_{t+1}^v = \mathbf{O}_t^v + \mu_t(\mathbf{Z}_{t+1}^v - \widehat{\mathbf{Z}}_{t+1} - \mathbf{E}_{2,t+1}^v), \tag{5}$$

we obtain

$$\mathbf{O}_{t+1}^v(:,i) \in \partial\lambda_2\|\mathbf{E}_{2,t+1}^v(:,i)\|_2. \tag{6}$$

Analogous to (4), $\{\mathbf{O}_t^v\}$ is bounded with $\|\mathbf{O}_t^v(:,i)\|_2 \leq \lambda_2$.

② For the $\mathcal{G}_{k,t+1}$ subproblem:

$$\min_{\mathcal{G}_k} \quad \lambda_2\|\mathcal{G}_k\|_{\circledS p}^p + \frac{\theta_t}{2}\left\|\nabla_k(\mathcal{A}_{t+1}) - \frac{\mathcal{N}_t^k}{\theta_t} - \mathcal{G}_k\right\|_{\mathrm{F}}^2, \tag{7}$$

the first-order optimality condition gives

$$0 \in \lambda_2\partial\|\mathcal{G}_{k,t+1}\|_{\circledS p}^p - \theta_t\left(\nabla_k(\mathcal{A}_{t+1}) - \frac{\mathcal{N}_t^k}{\theta_t} - \mathcal{G}_{k,t+1}\right). \tag{8}$$

Using the update rule

$$\mathcal{N}_{t+1}^k = \mathcal{N}_t^k + \theta_t(\mathcal{G}_{k,t+1} - \nabla_k(\mathcal{A}_{t+1})), \tag{9}$$

we derive

$$0 \in \lambda_2\partial\|\mathcal{G}_{k,t+1}\|_{\circledS p}^p - \mathcal{N}_{t+1}^k. \tag{10}$$

By Lemma 1, $\|\partial\|\mathcal{G}_{k,t+1}\|_{\circledS p}^p\|_{\mathrm{F}}$ is bounded, so $\{\mathcal{N}_t^k\}$ is bounded.

③ For the $\mathcal{A}_{t+1}$ subproblem:

$$\min_{\mathcal{A}} \sum_k^{\Gamma} \left\|\mathcal{G}_{k,t+1} - \nabla_k(\mathcal{A}) + \frac{\mathcal{N}_t^k}{\theta_t}\right\|_{\mathrm{F}}^2 + \left\|\mathcal{Z}_{t+1} - \mathcal{A} + \frac{\mathcal{C}_t}{\theta_t}\right\|_{\mathrm{F}}^2, \tag{11}$$

the first-order optimality condition yields

$$\sum_k^{\Gamma} \nabla_k^{\mathsf{T}}\left(\mathcal{G}_{k,t+1} - \nabla_k(\mathcal{A}_{t+1}) + \frac{\mathcal{N}_t^k}{\theta_t}\right) = \left(\mathcal{Z}_{t+1} - \mathcal{A}_{t+1} + \frac{\mathcal{C}_t}{\theta_t}\right). \tag{12}$$

Combining (9) with the update

$$\mathcal{C}_{t+1} = \mathcal{C}_t + \theta_t(\mathcal{Z}_{t+1} - \mathcal{A}_{t+1}), \tag{13}$$

we obtain

$$\mathcal{C}_{t+1} = \sum_k^{\Gamma} \nabla_k^{\mathsf{T}}(\mathcal{N}_{t+1}^k). \tag{14}$$

By Lemma 2 and the boundedness of $\mathcal{N}_{t+1}^k$, $\{\mathcal{C}_t\}$ is bounded.

Define $\mathcal{J}_t = \Phi(\mathbf{J}_t^1, \cdots, \mathbf{J}_t^V)$ and $\mathcal{O}_t = \Phi(\mathbf{O}_t^1, \cdots, \mathbf{O}_t^V)$. The Lagrangian satisfies

$$\begin{aligned}
&\mathcal{L}(\mathcal{Z}_t, \mathbf{E}_{1,t}^v, \mathbf{E}_{2,t}^v, \mathcal{J}_t, \mathcal{O}_t, \mathcal{N}_t^k, \mathcal{G}_{k,t}, \mathcal{C}_t, \mathcal{A}_t, \mu_t, \theta_t) \\
&\leq \mathcal{L}(\mathcal{Z}_t, \mathbf{E}_{1,t}^v, \mathbf{E}_{2,t}^v, \mathcal{J}_{t-1}, \mathcal{O}_{t-1}, \mathcal{N}_{t-1}^k, \mathcal{G}_{k,t-1}, \mathcal{C}_{t-1}, \mathcal{A}_t, \mu_t, \theta_t) \\
&\quad + \frac{\mu_t + \mu_{t-1}}{2\mu_{t-1}^2}\|\mathcal{J}_t - \mathcal{J}_{t-1}\|_{\mathrm{F}}^2 + \frac{\|\mathcal{J}_t\|_{\mathrm{F}}^2}{2\mu_t} - \frac{\|\mathcal{J}_{t-1}\|_{\mathrm{F}}^2}{2\mu_{t-1}} \\
&\quad + \frac{\mu_t + \mu_{t-1}}{2\mu_{t-1}^2}\|\mathcal{O}_t - \mathcal{O}_{t-1}\|_{\mathrm{F}}^2 + \frac{\|\mathcal{O}_t\|_{\mathrm{F}}^2}{2\mu_t} - \frac{\|\mathcal{O}_{t-1}\|_{\mathrm{F}}^2}{2\mu_{t-1}} \\
&\quad + \sum_k^{\Gamma}\left(\frac{\theta_t + \theta_{t-1}}{2\theta_{t-1}^2}\|\mathcal{N}_t^k - \mathcal{N}_{t-1}^k\|_{\mathrm{F}}^2 + \frac{\|\mathcal{N}_t^k\|_{\mathrm{F}}^2}{2\theta_t} - \frac{\|\mathcal{N}_{t-1}^k\|_{\mathrm{F}}^2}{2\theta_{t-1}}\right) \\
&\quad + \frac{\theta_t + \theta_{t-1}}{2\theta_{t-1}^2}\|\mathcal{C}_t - \mathcal{C}_{t-1}\|_{\mathrm{F}}^2 + \frac{\|\mathcal{C}_t\|_{\mathrm{F}}^2}{2\theta_t} - \frac{\|\mathcal{C}_{t-1}\|_{\mathrm{F}}^2}{2\theta_{t-1}}.
\end{aligned} \tag{15}$$

Summing both sides of (15) gives

$$\begin{aligned}
&\mathcal{L}(\mathcal{Z}_n, \mathbf{E}_{1,n}^v, \mathbf{E}_{2,n}^v, \mathcal{J}_n, \mathcal{O}_n, \mathcal{N}_n^k, \mathcal{G}_{k,n}, \mathcal{C}_n, \mathcal{A}_n, \mu_n, \theta_n) \\
&\le \mathcal{L}(\mathcal{Z}_0, \mathbf{E}_{1,0}^v, \mathbf{E}_{2,0}^v, \mathcal{J}_0, \mathcal{O}_0, \mathcal{N}_0^k, \mathcal{G}_{k,0}, \mathcal{C}_0, \mathcal{A}_0, \mu_0, \theta_0) \\
&\quad + \frac{\|\mathcal{J}_n\|_{\mathrm{F}}^2}{2\mu_n} - \frac{\|\mathcal{J}_0\|_{\mathrm{F}}^2}{2\mu_0} + \sum_{t=1}^{n} \frac{\mu_t + \mu_{t-1}}{2\mu_{t-1}^2} \|\mathcal{J}_t - \mathcal{J}_{t-1}\|_{\mathrm{F}}^2 \\
&\quad + \frac{\|\mathcal{O}_n\|_{\mathrm{F}}^2}{2\mu_n} - \frac{\|\mathcal{O}_0\|_{\mathrm{F}}^2}{2\mu_0} + \sum_{t=1}^{n} \frac{\mu_t + \mu_{t-1}}{2\mu_{t-1}^2} \|\mathcal{O}_t - \mathcal{O}_{t-1}\|_{\mathrm{F}}^2 \\
&\quad + \sum_{k}^{\Gamma} \left[\frac{\|\mathcal{N}_n^k\|_{\mathrm{F}}^2}{2\theta_n} - \frac{\|\mathcal{N}_0^k\|_{\mathrm{F}}^2}{2\theta_0} + \sum_{t=1}^{n} \frac{\theta_t + \theta_{t-1}}{2\theta_{t-1}^2} \|\mathcal{N}_t^k - \mathcal{N}_{t-1}^k\|_{\mathrm{F}}^2 \right] \\
&\quad + \frac{\|\mathcal{C}_n\|_{\mathrm{F}}^2}{2\theta_n} - \frac{\|\mathcal{C}_0\|_{\mathrm{F}}^2}{2\theta_0} + \sum_{t=1}^{n} \frac{\theta_t + \theta_{t-1}}{2\theta_{t-1}^2} \|\mathcal{C}_t - \mathcal{C}_{t-1}\|_{\mathrm{F}}^2.
\end{aligned} \tag{16}$$

Given

$$\sum_{t=1}^{\infty} \frac{\mu_t + \mu_{t-1}}{2\mu_{t-1}^2} < \infty, \quad \sum_{t=1}^{\infty} \frac{\theta_t + \theta_{t-1}}{2\theta_{t-1}^2} < \infty, \tag{17}$$

and the boundedness of $\{\mathcal{J}_t, \mathcal{O}_t, \mathcal{N}_t^k, \mathcal{C}_t\}$, the right-hand side of (16) is finite. Thus $\mathcal{L}(\mathcal{Z}_n, \mathbf{E}_{1,n}^v, \mathbf{E}_{2,n}^v, \mathcal{J}_n, \mathcal{O}_n, \mathcal{N}_n^k, \mathcal{G}_{k,n}, \mathcal{C}_n, \mathcal{A}_n, \mu_n, \theta_n)$ is bounded. Observing that

$$\begin{aligned}
&\mathcal{L}(\mathcal{Z}_n, \mathbf{E}_{1,n}^v, \mathbf{E}_{2,n}^v, \mathcal{J}_n, \mathcal{O}_n, \mathcal{N}_n^k, \mathcal{G}_{k,n}, \mathcal{C}_n, \mathcal{A}_n, \mu_n, \theta_n) \\
&= \lambda_1 \sum_{k}^{\Gamma} \|\mathcal{G}_{k,n}\|_{\circledS p}^p + \lambda_2 \sum_{v=1}^{V} \|\mathbf{E}_n^v\|_{2,1} \\
&\quad + \sum_{v=1}^{V} \frac{\mu_n}{2} \left\| \mathbf{X}^v - \mathbf{X}^v \mathbf{Z}_n^v - \mathbf{E}_{1,n}^v + \frac{\mathbf{J}_n^v}{\mu_n} \right\|_{\mathrm{F}}^2 \\
&\quad + \sum_{v=1}^{V} \frac{\mu_n}{2} \left\| \mathbf{Z}_n^v - \widehat{\mathbf{Z}}_n - \mathbf{E}_{2,n}^v + \frac{\mathbf{O}_n^v}{\mu_n} \right\|_{\mathrm{F}}^2 \\
&\quad + \sum_{k}^{\Gamma} \frac{\theta_n}{2} \left\| \mathcal{G}_{k,n} - \nabla_k(\mathcal{A}_n) + \frac{\mathcal{N}_n^k}{\theta_n} \right\|_{\mathrm{F}}^2 \\
&\quad + \frac{\theta_n}{2} \left\| \mathcal{Z}_n - \mathcal{A}_n + \frac{\mathcal{C}_n}{\theta_n} \right\|_{\mathrm{F}}^2,
\end{aligned} \tag{18}$$

where all terms are non-negative, each term is bounded. Specifically, boundedness of $\|\mathcal{G}_{k,n}\|_{\circledS p}^p$ implies $\|\mathcal{G}_{k,t}\|_{\mathrm{F}}^2$ is bounded, and similarly $\{\mathbf{E}_{1,t}^v, \mathbf{E}_{2,t}^v\}$ is bounded.

To establish boundedness of $\mathcal{Z}_t$, factorize $\mathbf{Z}_n^v = \mathbf{H}_n^v \bar{\mathbf{Z}}_n^v$ where $\mathbf{H}_n^v$ is an orthonormal basis for the column space of $(\mathbf{X}^v)^{\mathsf{T}}$. Defining $\mathbf{R}_n^v = \|\mathbf{X}^v - \mathbf{X}^v \mathbf{Z}_n^v - \mathbf{E}_{1,n}^v + \mathbf{J}_n^v/\mu_n\|_{\mathrm{F}}^2$ and using the Moore–Penrose inverse $(\mathbf{X}^v)^+$, we derive

$$\mathbf{Z}_n^v = \mathbf{I}^v - (\mathbf{X}^v)^+ \mathbf{R}_n^v - (\mathbf{X}^v)^+ \mathbf{E}_{1,n}^v - (\mathbf{X}^v)^+ \frac{\mathbf{J}_n^v}{\mu_n},$$

yielding

$$\begin{aligned}
\|\mathbf{Z}_n^v\|_{\mathrm{F}} &\le \sqrt{N} + \|(\mathbf{X}^v)^+\|_{\mathrm{F}} \left(\|\mathbf{R}_n^v\|_{\mathrm{F}} + \|\mathbf{E}_{1,n}^v\|_{\mathrm{F}} + \left\| \frac{\mathbf{J}_n^v}{\mu_n} \right\|_{\mathrm{F}} \right) \\
&< \infty,
\end{aligned} \tag{19}$$

under boundedness of $\mathbf{R}_n^v$, $\mathbf{X}^v$, $\mathbf{E}_{1,n}^v$, and $\mathbf{J}_n^v/\mu_n$. Thus $\{\mathbf{Z}_t^v, \widehat{\mathbf{Z}}_t\}$ is bounded. Since $(\mathcal{C}_t - \mathcal{C}_{t-1})/\theta_{t-1} = \mathcal{Z}_t - \mathcal{A}_t$, $\{\mathcal{A}_t\}$ is bounded.
Hence $\{P_t\}$ is bounded.

Proof of (2). By the Bolzano–Weierstrass theorem, the bounded sequence $\{P_t\}$ has an accumulation point $P^* = \{\mathcal{Z}^*, \mathbf{E}_1^{v,*}, \mathbf{E}_2^{v,*}, \mathcal{J}^*, \mathcal{O}^*, \mathcal{N}^{k,*}, \mathcal{C}^*, \mathcal{A}^*\}$. From update rules:

$$\begin{aligned}
\mathbf{J}_{t+1}^v &= \mathbf{J}_t^v + \mu_t(\mathbf{X}^v - \mathbf{X}^v \mathbf{Z}_t^v - \mathbf{E}_{1,t}^v); \\
\mathbf{O}_{t+1}^v &= \mathbf{O}_t^v + \mu_t(\mathbf{Z}_t^v - \widehat{\mathbf{Z}}_t - \mathbf{E}_{2,t}^v); \\
\mathcal{N}_{t+1}^k &= \mathcal{N}_t^k + \theta_t(\mathcal{G}_{k,t} - \nabla_k(\mathcal{A}_t)); \\
\mathcal{C}_{t+1} &= \mathcal{C}_t + \theta_t(\mathcal{Z}_t - \mathcal{A}_t),
\end{aligned} \tag{20}$$

we obtain at P^*:

$$\begin{aligned}
\mathbf{X}^v &= \mathbf{X}^v \mathbf{Z}^{v,*} - \mathbf{E}_1^{v,*}; \\
\mathbf{Z}^{v,*} &= \widehat{\mathbf{Z}}^* + \mathbf{E}_2^{v,*}; \\
\mathcal{G}_k^* &= \nabla_k(\mathcal{A}^*); \\
\mathcal{Z}^* &= \mathcal{A}^*.
\end{aligned} \tag{21}$$

From $\mathbf{E}_1^v$ and $\mathbf{E}_2^v$ subproblems:

$$\begin{aligned}
\mathbf{J}_{t+1}^v \in \partial\lambda_2 \|\mathbf{E}_{1,t+1}^v\|_{2,1} &\Rightarrow \mathbf{J}^{v*} \in \partial\lambda_2 \|\mathbf{E}_1^{v,*}\|_{2,1}; \\
\mathbf{O}_{t+1}^v \in \partial\lambda_2 \|\mathbf{E}_{2,t+1}^v\|_{2,1} &\Rightarrow \mathbf{O}^{v*} \in \partial\lambda_2 \|\mathbf{E}_2^{v,*}\|_{2,1}.
\end{aligned} \tag{22}$$

From tensor subproblems:

$$\begin{aligned}
\mathcal{N}_{t+1}^k \in \lambda_2 \partial \|\mathcal{G}_{k,t+1}\|_{\circledS p}^p &\Rightarrow \mathcal{N}^{k,*} \in \lambda_2 \partial \|\mathcal{G}_k^*\|_{\circledS p}^p, \\
\mathcal{C}^* = \sum_{k}^{\Gamma} \nabla_k^{\mathsf{T}}(\mathcal{N}^{k,*}) \quad &(\text{from } \mathcal{C}_{t+1} \text{ update}).
\end{aligned} \tag{23}$$

P^* satisfies the KKT conditions.
This is the completed proof.

Corollary 1 (Geometric Convergence)**.** *Assume multipliers are bounded: $\|\mathbf{J}^v\|_F \le M_J$, $\|\mathbf{O}^v\|_F \le M_O$, $\|\mathcal{N}^k\|_F \le M_N$, $\|\mathcal{C}\|_F \le M_C$. With adaptive penalties $\mu_t = \mu_0 \rho^t$, $\theta_t = \theta_0 \rho^t$ ($\rho > 1$), the primal residual decays linearly:*

$$R_t \le C\beta^t,$$

where $C = \max\left(\frac{2\rho M_J}{\mu_0}, \frac{2\rho M_O}{\mu_0}, \frac{2\rho M_N}{\theta_0}, \frac{2\rho M_C}{\theta_0}\right)$, $\beta = \rho^{-1}$. This implies $\mathcal{O}(\ln \epsilon^{-1})$ iteration complexity.

Proof From multiplier updates:

$$\begin{aligned}
\|\mathbf{X}^v - \mathbf{X}^v \mathbf{Z}_t^v - \mathbf{E}_{1,t}^v\|_F &= \mu_{t-1}^{-1} \|\mathbf{J}_t^v - \mathbf{J}_{t-1}^v\|_F \le 2M_J \mu_{t-1}^{-1} \\
\|\mathbf{Z}_t^v - \widehat{\mathbf{Z}}_t - \mathbf{E}_{2,t}^v\|_F &= \mu_{t-1}^{-1} \|\mathbf{O}_t^v - \mathbf{O}_{t-1}^v\|_F \le 2M_O \mu_{t-1}^{-1} \\
\|\mathcal{G}_k - \nabla_k(\mathcal{A}_t)\|_F &= \theta_{t-1}^{-1} \|\mathcal{N}_t^k - \mathcal{N}_{t-1}^k\|_F \le 2M_N \theta_{t-1}^{-1} \\
\|\mathcal{Z}_t - \mathcal{A}_t\|_F &= \theta_{t-1}^{-1} \|\mathcal{C}_t - \mathcal{C}_{t-1}\|_F \le 2M_C \theta_{t-1}^{-1}
\end{aligned}$$

Substituting $\mu_{t-1} = \mu_0 \rho^{t-1}$, $\theta_{t-1} = \theta_0 \rho^{t-1}$ yields:

$$\mu_{t-1}^{-1} = \tfrac{\rho}{\mu_0} \beta^t, \quad \theta_{k-1}^{-1} = \tfrac{\rho}{\theta_0} \beta^k$$

Thus $R_t = \max(\text{residuals}) \le C\beta^t$ with C as defined. Solving $C\beta^t \le \epsilon$ gives $t \ge \log_\rho(\epsilon/C) = \mathcal{O}(\ln \epsilon^{-1})$.